

Experiment 1: Microbial Culture Media Preparation,

Introduction

Media (singular, medium) are the nutrient preparations used to grow microorganisms in the laboratory. Depending on the chemical composition a medium can be categorized as **synthetic** and **complex** media. Synthetic media are chemically defined in which the concentration of each nutrient is known. Complex media are made of components rich in nutrients but the exact composition of each nutrient is unknown. For growing microorganisms complex media are suffice and advantageous. They contain soluble digests of plant or animal tissue, such as casein, beef extract, yeast extract, soybeans, peptones etc. Media can be prepared either in liquid or solid forms. Based on the intended use, its contents may vary as in Selective, differential and enrichment media. **Selective media** contains compounds that selectively allows the growth of some microorganism but inhibit others. **Differential media** contains some sort of indicator, or a dye, that distinguishes between species of bacteria, some of which may carry out a particular reaction while others do not. **Enriched medium** is one to which additional nutrients such as serum or whole blood is added to culture some fastidious organisms that have stringent nutritional requirements.

Liquid media such as nutrient broth, tryptic soy broth, or brain-heart infusion broth, can be used to propagate large numbers of microorganisms in fermentation studies and for various biochemical tests.

Solid media contains a solidifying agent called **agar** to the liquid media e.g., nutrient agar, blood agar. Solid media are used (1) to allow growth of visible and isolated masses of microorganisms called colonies, (2) obtaining pure culture, (3) storage of cultures, and (4) to observe specific biochemical reactions.

Objectives:

- Use aseptic procedures to prepare and transfer culture media.
- Describe the concept of sterility and disinfection
- Enable students use autoclave correctly and safely
- Describe how various media, supplies, and equipment can be sterilized

Materials and Equipments:

Nutrient agar and Nutrient broth powder	Distilled water
Analytical balance	Hot plates
Aluminum foil	Measuring cylinder
Autoclave	Spatula
Petri dishes	Boiling water bath
Erlenmeyer flask	Test tubes
Test tube rack	Bunsen burner

Procedure:

- 1) Rehydrate the powder according to manufacturer's instructions. Ensure ingredients are completely dissolved by boiling if necessary
- 2) Dispense in to test tubes to prepare broth or slant agar which then be placed in test tube rack to be placed in the autoclave.
- 3) Sterilize the media using autoclave at 121°C, 15 lbs pressure for 15 minutes.
- 4) Sterilized media is then transferred to a 48°C to 50°C
- 5) **Pouring into a Petri plate**

Discussion Questions:

1. Why are culture media sterilized before use?
2. Why is culture medium cooled to about 48°C to 50°C before it is poured into Petri plates?
3. Why are Petri plates inverted after they cool?

Experiment 2: Sterilization and Disinfection

Sterilization is the complete destruction of all microorganisms including endospores, and this can be achieved by use of heat, chemicals, and radiation and filtration methods.

a. Autoclave: The most useful approach is **autoclaving**, in which items are sterilized by exposure to steam at 121°C and 15 lbs of pressure for 15 minutes or longer, depending on the nature of the item.

b. Dry Heat Sterilization: Hot-Air Oven at 160°C for two or more hours.

c. Red Heat: Inoculating wires or loops are sterilized by holding them in a Bunsen burner flame until they are red hot..

d. Radiation: Gamma and UV radiations can be employed for sterilization of heat-sensitive materials. UV light cannot penetrate materials so is used mainly for surface treatments e.g. UV lamps fitted in Biological safety cabinets and Laminar flow hoods to sterilize the inside air.

e. Filtration: Membrane Filters act by screening out particles. Membrane filtration is usually employed for heat-sensitive liquids and media that will not withstand heating at 121°C.

Disinfection

Disinfection is the destruction of most pathogenic microorganism (except endospores), so they are no longer a health hazard. It is usually achieved by the use of chemicals. Disinfectants at specified working strengths are used for specific purposes.

Chemical agents commonly used as disinfectants:

- **Sodium Hypochlorite (5000 - 6000 ppm)** (household bleach) - to sanitize glasswares
- **Ethanol 70% (v/v)** industrial methylated spirit
- **Iodine tincture (2% in 70% alcohol) and Iodophores**
- **Phenolics**
- **Heavy Metals (AgNO_3 & AgCl_2)**
- **Soaps and detergents (Quaternary ammonium compounds)**

When preparing working strength solutions from stock and dealing with powder form, wear eye protection and gloves to avoid irritant or harmful effects. Disinfectants for use at working strength should be freshly prepared from full strength stock or powder form.